

Project 02: RAG-Based Question Answering AI Studio

Enterprise Retrieval-Augmented Generation Grounded in Corpus Context

1. Executive Overview

The RAG QA Engine provides grounded document intelligence. It ingests custom knowledge bases, performs automated text chunking with overlap (250 chars, 30 overlap), indexes chunks via TF-IDF cosine similarity, and synthesizes answers with source citations.

2. System Architecture & Workflow Pipeline

- Document Ingestion -> DocumentChunker (250 chars, 30 overlap)
- SimpleVectorStore TF-IDF Cosine Similarity Indexing
- User Question -> Top-K Chunk Retrieval & Grounded Context Assembly
- Google Gemini 2.5 Flash Grounded Synthesis -> Answer with [Source | Chunk ID] Citations

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves RAG AI Studio web UI
GET	/api/status	Returns vector store status, chunk counts, and indexed sources
GET	/api/documents	Lists indexed documents and chunk previews
POST	/api/ingest	Ingests new text content into vector store
POST	/api/query	Executes RAG query and returns grounded answer with citations

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 02_rag_qa/src && python -m uvicorn server:app --host 127.0.0.1 --port 8002
```

Automated Test Suite Execution:

```
python -m pytest 02_rag_qa/tests/
```